Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title	**ELE827**
Course Name	**CMOS Mixed Signal**
Semester/Year	**F2026**

Instructor	**Dr Fei Yuan**

Lab Number	**Lab 3**
Lab Name	**Comparator**

Submission Date	
Due Date	

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Table of Contents:

Schematic - p3
Input offset Voltage - p4
Input offset Voltage compensation - p6
Kickback - p7
Delay - p7
Conclusion - p8

List of Figures

Figure 1: Comparator Schematic
Figure 2: Comparator Symbol
Figure 3: DC Sweep of C+ (Flatline at > 700 mV)
Figure 4: DC Sweep of C- (Flatline at > 700 mV)
Figure 5: Mismatch Sweep of C+ (Flatline at < 500 nm)
Figure 6: Mismatch Sweep of C- (Flatline at > 500 nm)
Figure 7: Thermometer Capacitor Sweep for C+ (Calibration Boundary Occurs at 4 Caps)
Figure 8: Thermometer Capacitor Sweep for C- (Calibration Boundary Occurs at 3 Caps from 8)
Figure 9: Kickback-Induced Output Flipping Between 2 nV and 3 nV
Figure 10: Comparator Delay vs Input Differential Voltage

List of Tables

Table 1: Component Values
Table 2: Delay vs V

List of Abbreviations

ADE — Analog Design Environment
ADE XL — Analog Design Environment Explorer
TB — Testbench
TRAN — Transient Analysis
DC — Direct Current (DC analysis / operating point)
PVT — Process / Voltage / Temperature
TT / FF / SS — Typical / Fast / Slow process corners
VDD — Supply Voltage
GND — Ground
VIN — Input Voltage
VOUT — Output Voltage

Schematic:

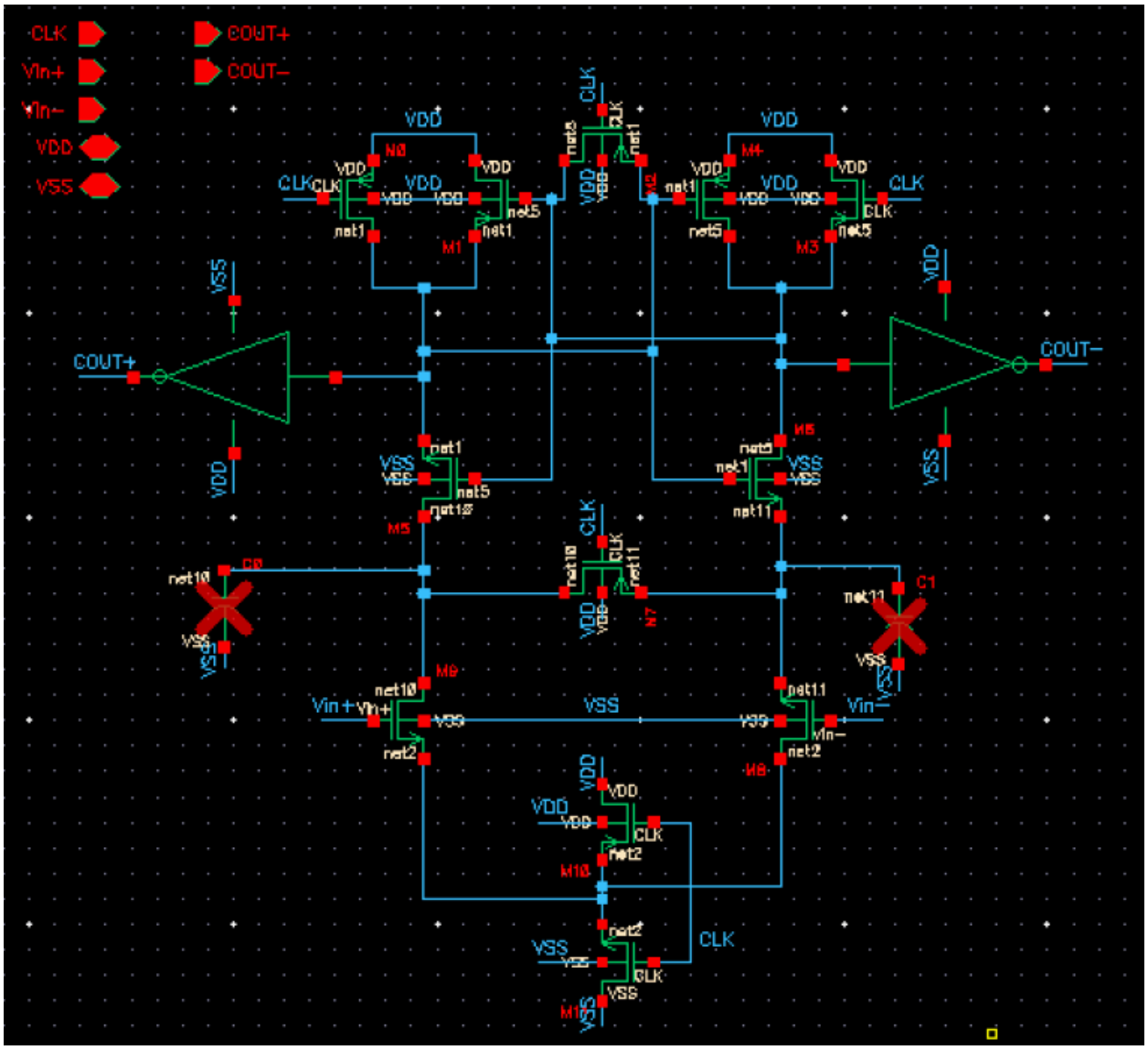


Figure 1: Comparator Schematic

Component	Width
All PMOS	900 n
Input NMOS	500 n
Other NMOS	300 n

Table 1: Component Values

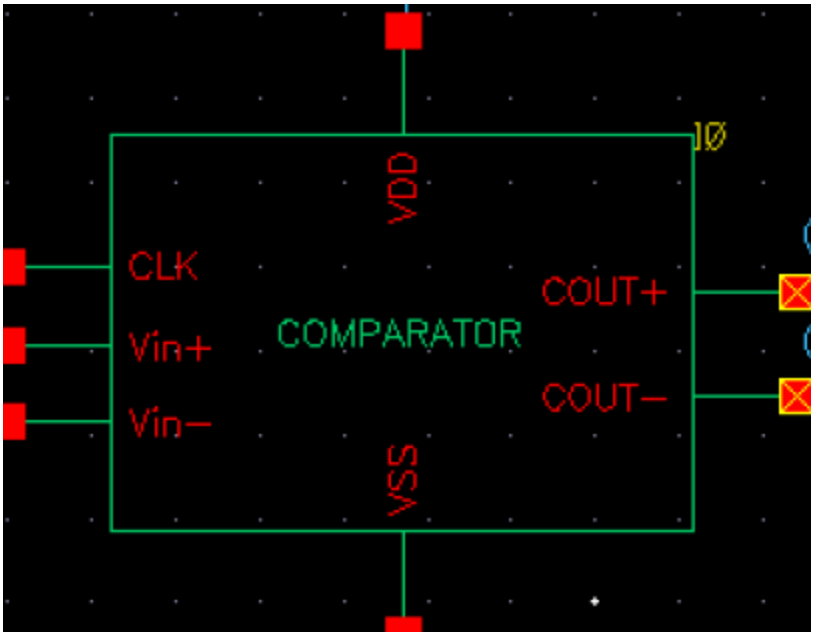


Figure 2: Comparator Symbol

Input offset voltage:

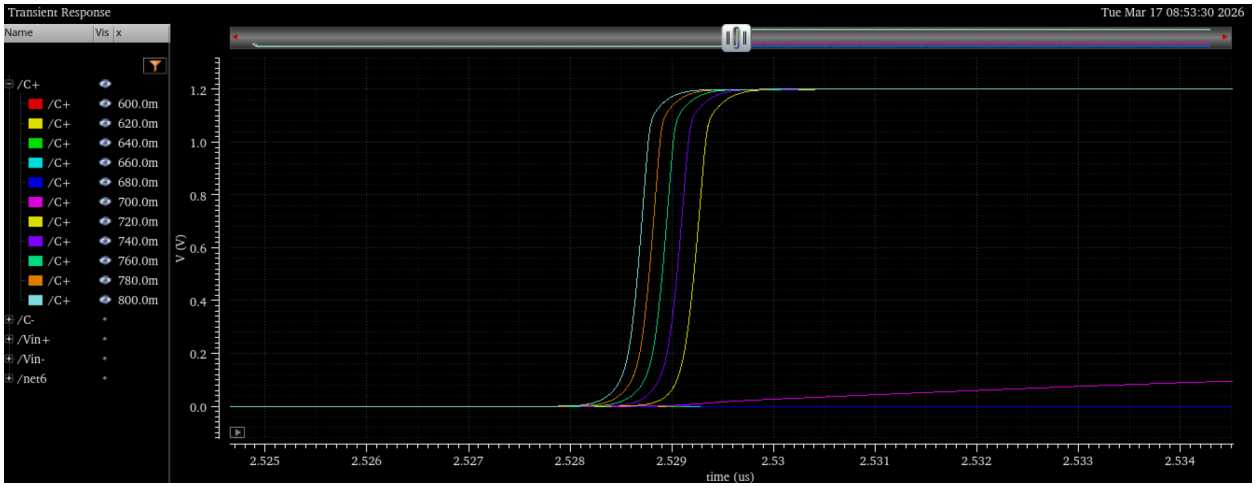


Figure 3: DC Sweep of C+ (Flatline at > 700 mV)

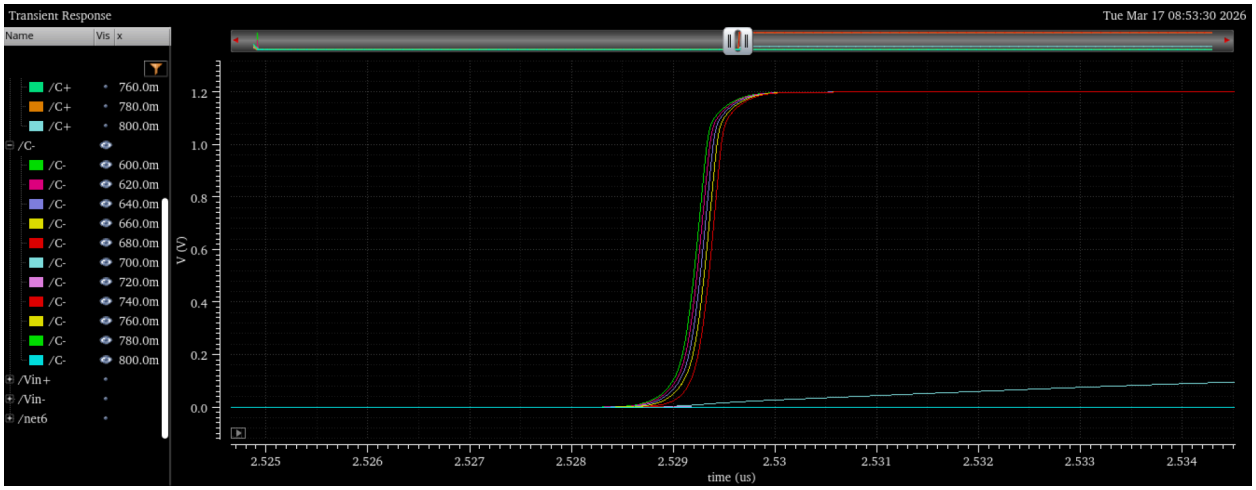


Figure 4: DC Sweep of C- (Flatline at > 700mV)

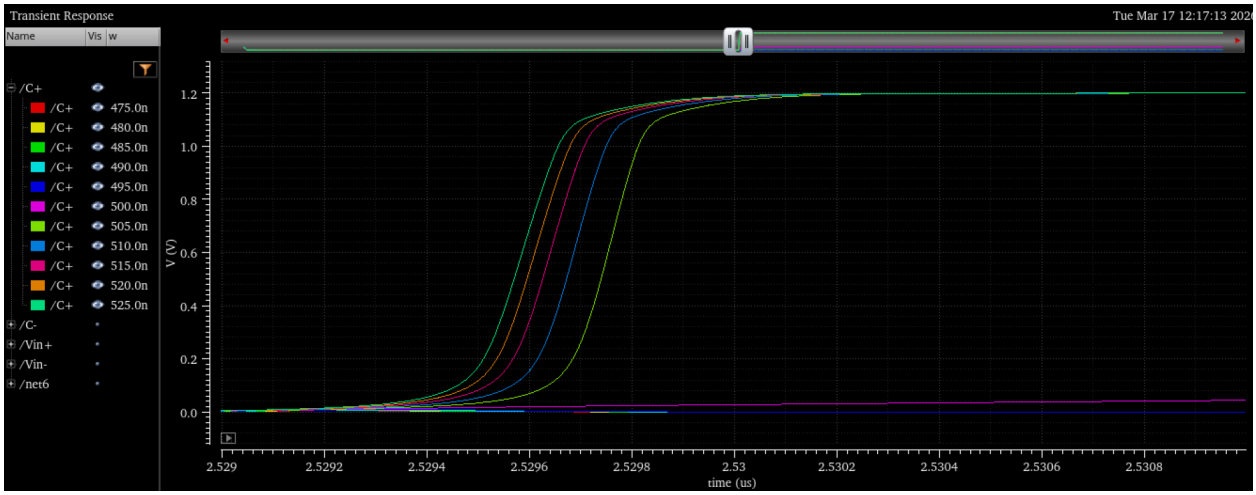


Figure 5: Mismatch Sweep of C+(Flatline at < 500n)

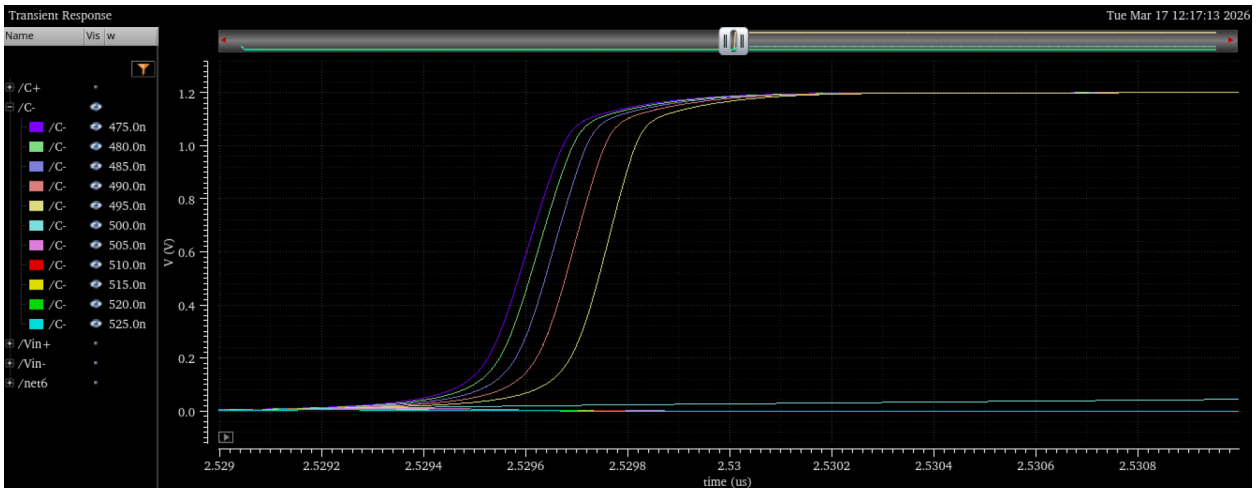


Figure 6: Mismatch Sweep of C- (Flatline at > 500n)

Input offset voltage compensation:

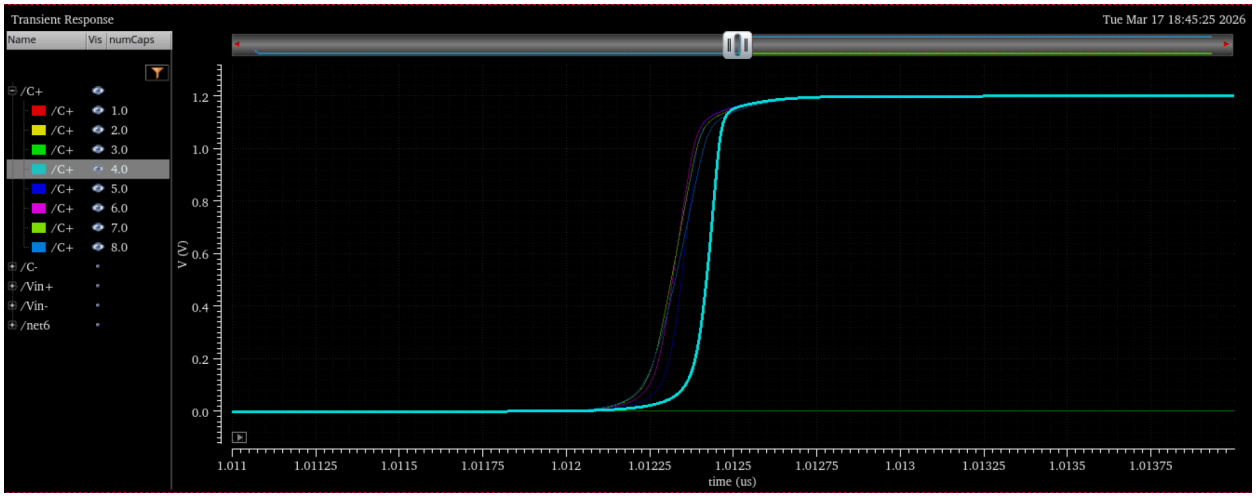


Figure 7: Thermometer Capacitor Sweep for C+ (Calibration Boundary Occurs at 4 Caps)

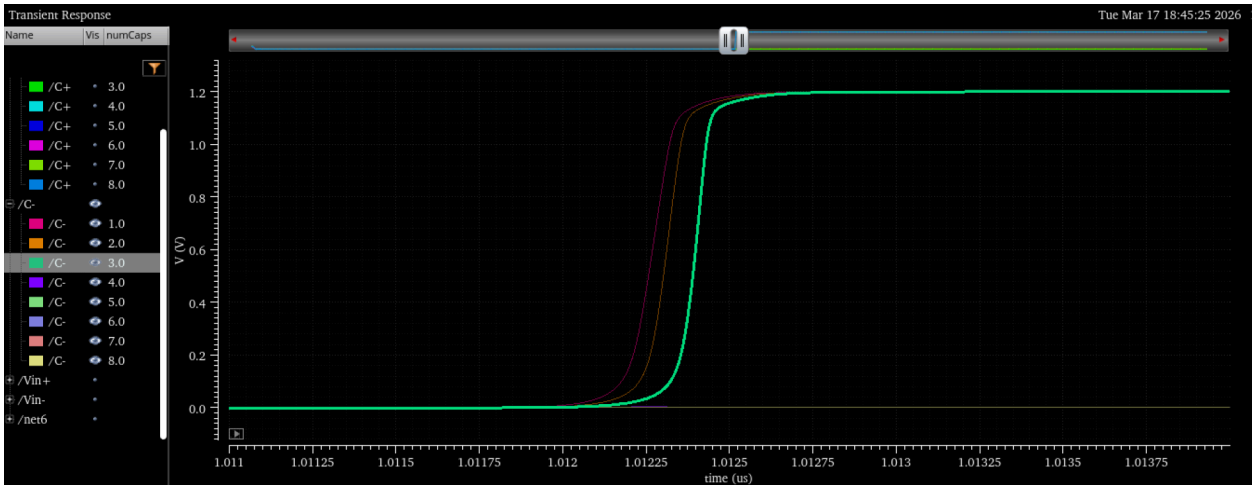


Figure 8: Thermometer Capacitor Sweep for C- (Occurs at 3 Caps starting from 8)
N = 4 - 3, Resolution = Vos/2

Kickback:

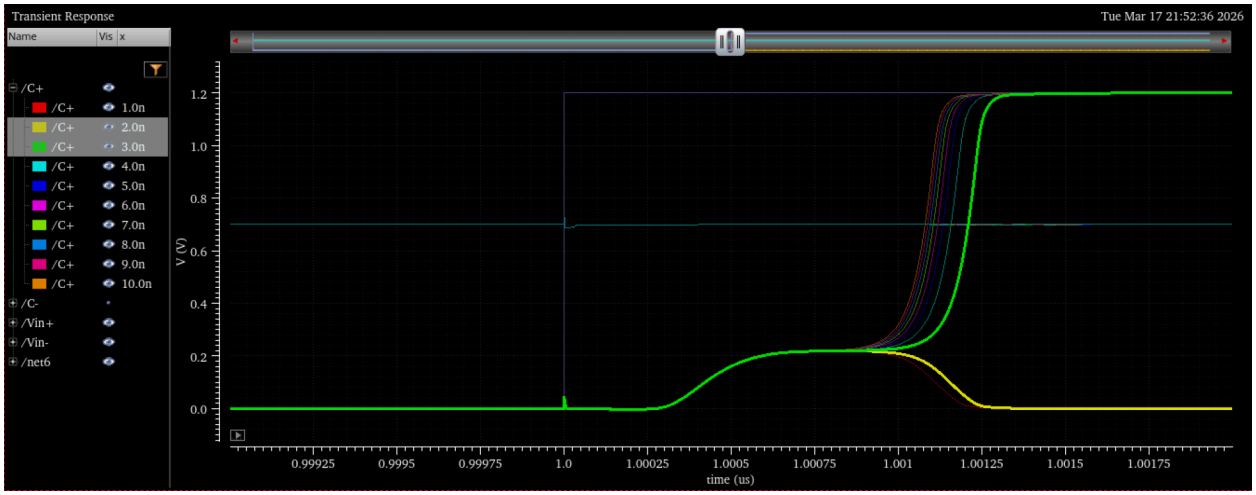


Figure 9: Kickback Flipping between $\triangle V = 2nV \; and \; 3nV$
(Note: Vin+ = 700mV + $\triangle V$, Vin- = 700mV - $\triangle V$)

Delay:

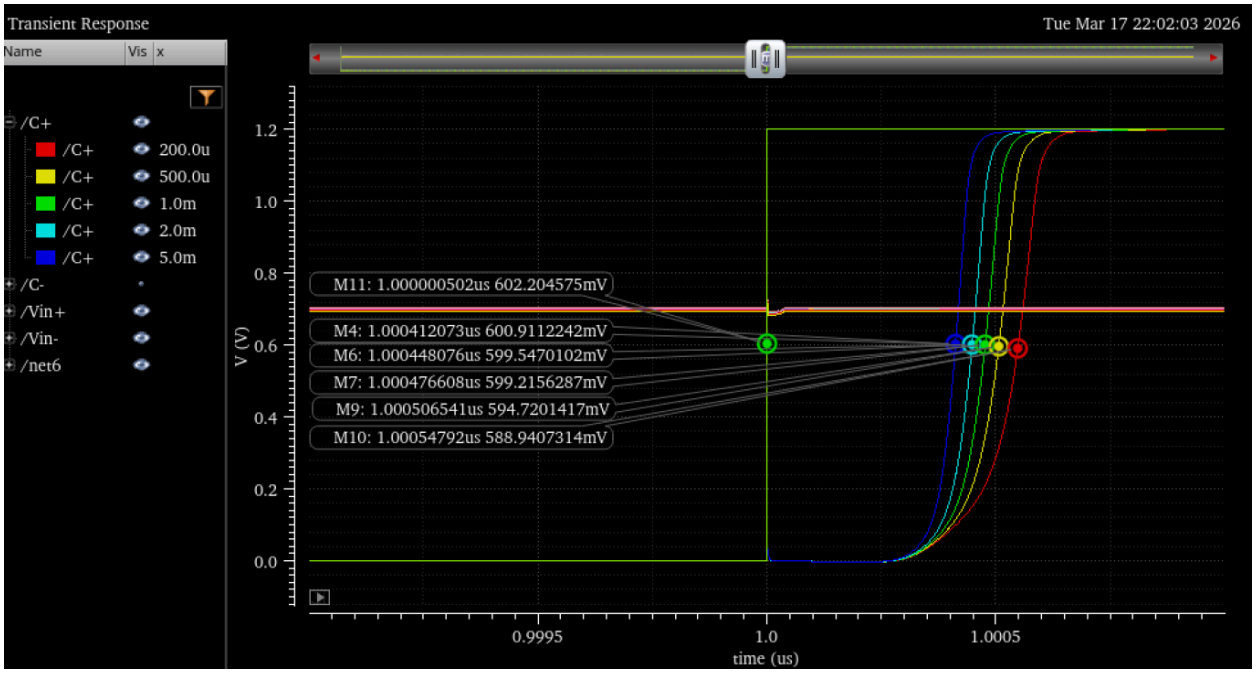


Figure 10: Delay with respect to $\triangle V$

$\triangle V$ (mV)	Clock Time (us)	Time at 0.6V Crossing (us)	Delay (ps)
0.2	1.00000052	1.00054792	547.4

0.5	1.00000052	1.00050645	505.9
1	1.00000052	1.00047660	476.1
2	1.00000052	1.00044808	447.6
5	1.00000052	1.00041207	411.5

Table 2: Delay vs $\triangle V$

Conclusion

In this lab, the comparator component of the SAR ADC was analyzed in terms of its threshold behavior, input offset voltage, kickback, and delay. The results show that mismatch introduces offset, which can be seen in **Figure 5 & 6** where the transistor width was swept from 475n to 525n. This error, however, can be compensated using a thermometer coded capacitor array, which allows tuning of the capacitance at the comparator drain nodes to compensate for mismatch error. Kickback is also explored as the input voltages are swept with a small difference to determine the voltage at which the comparator produces an error. Larger differences between Vin+ and Vin- results in faster decision time which can be seen in **Figure 10** and **Table 2**.